

MiniDxO: The Transparent AI Diagnostician

A lightweight, multi-agent web app for trustworthy medical reasoning

Team AION

AION — MiniDxO Project

Hackathon Pitch

Quick Overview

- **Problem:** Current "black-box" medical AIs give opaque, one-shot answers.
- **Solution:** MiniDxO — an interactive, transparent AI that simulates clinical reasoning and cites verified sources.
- **Goal:** Build user trust via visible reasoning, live questioning, and source citations.

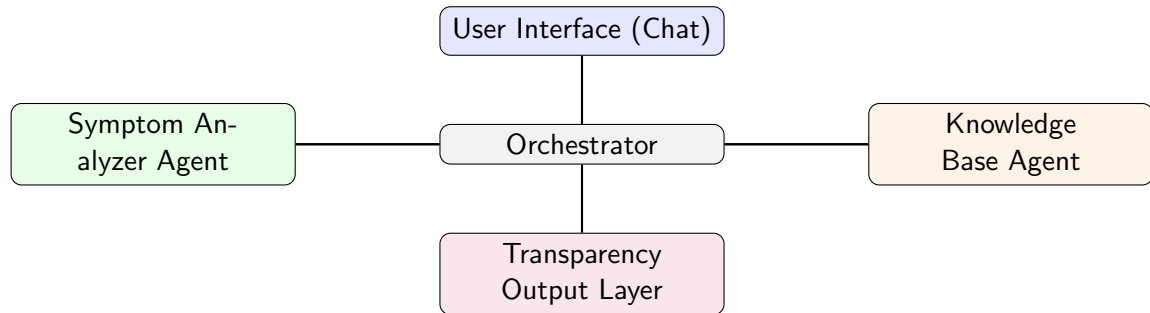
User Journey Example

- 1 User: "I have fever and sore throat."
- 2 MiniDxO: "Do you have white patches in your throat?" (explains why)
- 3 User: "Yes."
- 4 MiniDxO: Displays reasoning chain, cites Mayo Clinic, outputs likely diagnosis (Strep Throat, Confidence 78%).

Core Pillars

- **Interactive Questioning** — adaptive, context-aware follow-ups.
- **Cited Evidence** — trusted medical sources (Mayo Clinic, NIH, MedlinePlus).
- **Transparency Layer** — shows reasoning trace, confidence updates, and provenance.

Proposed Architecture



Why Multi-Agent?

- **Specialization:** Separate agents for triage, questioning, verification.
- **Transparency:** Each agent's reasoning trace is viewable.
- **Modularity:** Components can be updated independently.

Frontend (Web) — Design Highlights

- Built with **React** or Svelte for reactive chat-based UI.
- Two panels — *Conversation* and *Reasoning Log*.
- Confidence updates: “Confidence: 35% → 78%”.
- Responsive design for mobile and web.

Backend Tech Stack

- **API Layer:** Node.js/Express or FastAPI for orchestration.
- **Agents:** Python microservices (small LLM calls + rule-based logic).
- **Knowledge Store:** Indexed articles using SQLite/ElasticSearch.
- **Security:** Basic consent + privacy (HIPAA-aligned for scaling).
- **Deployment:** Docker + Cloud Run / Vercel for scalability.

Minimum Viable Product (MVP)

- 1 Chat UI with interactive reasoning trace.
- 2 Core agents: Symptom Analyzer, Question Generator, Knowledge Fetcher.
- 3 Local medical database (few verified sources).
- 4 Confidence scoring + citation snippets.

Demo Script (2–3 Minutes)

- ➊ Input: “Fever + sore throat.”
- ➋ MiniDxO asks a follow-up question.
- ➌ Show confidence and citation updates.
- ➍ End: Explain reasoning path and transparency.

Safety, Ethics, and Limitations

- **Informational Only** — not a clinical replacement.
- Cite every output source with confidence bounds.
- Include disclaimers + user consent logging.

Timeline Team Roles

Role	Responsibility
Frontend	Chat UI, Reasoning display
Backend	APIs, Orchestration logic
Data	Knowledge ingestion, citations
ML / Agents	Triage logic, follow-ups
Testing	Safety, usability checks

How to Present on Stage

- Problem (15–20s)
- Live Demo (60–90s)
- Architecture (30s)
- Future Work Ask (30s)

Next Steps

- Populate knowledge base with verified medical content.
- Implement transparent reasoning display.
- Test with common triage flows.

Thank You

Questions?

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